



Paper Trail

User Interface Design Document

Dr. Zihan Wu

Ben Yandell, Robert Kulow, Anthony Veilleux, Arius Ahmad, Vasu Patel

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V1.0

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1. Introduction

This is a capstone project commissioned by Dr. Zihan Wu. This project will fulfill the capstone requirement for a Computer Science Bachelor's degree for Ben Yandell, Robert Kulow, Anthony Veilleux, Arius Ahmad, and Vasu Patel. As a researcher, the client has expressed frustration with the difficulty that comes with maintaining a single “main” document where progress, notes, and references are collected. They noticed that Google Docs doesn't natively support project management functions. When they tried third-party applications like Notion and Asana, they found them very convoluted and unintuitive. This project aims to offer an intuitive plugin that works directly within Google Docs while writing. The client proposed a tagging system in which any paragraph that follows a special character (“#”, “@”, etc.) is treated as a note until a page break or another tag. Thus, differentiating notes from actual documentation. The client also requested multiple types of tags (type of note, project it's under, date, etc.) to allow for more targeted searching functionalities. Also, they requested a stretch goal: the ability to do quick commands with backslashes while typing in the document to bring up to-do tables and such on the fly.

1.1 Purpose of This Document

The purpose of this User Interface Design Document is to define the visual and interactive components of the PaperTrail plugin. This document gives a detailed description of the user interaction components, including navigation structure, interactive elements, and visual styling. It translates the System Requirements Specifications functional requirements into interface design that will guide front-end implementation. This document ensures that the product is intuitive and aligned with the goals and user needs of the PaperTrail plugin.

1.2 References

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2025) *PaperTrail System Requirements Specification*

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2025) *PaperTrail System Design Document*

Arius, A., Anthony, V., Ben, Y., Robert, K., Vasu, P. (2025) *PaperTrail User Interface Design Document*

2. User Interface Standards

This section defines the design standards that will maintain consistency throughout the Research Tag Manager system. The interface follows Google Docs design principles to ensure familiarity for researchers and seamless integration with existing document workflows.

2.1 Overall Screen Layout

The Research Tag Manager system utilizes a three-panel layout as illustrated in Figure 1.

Software User Interface:

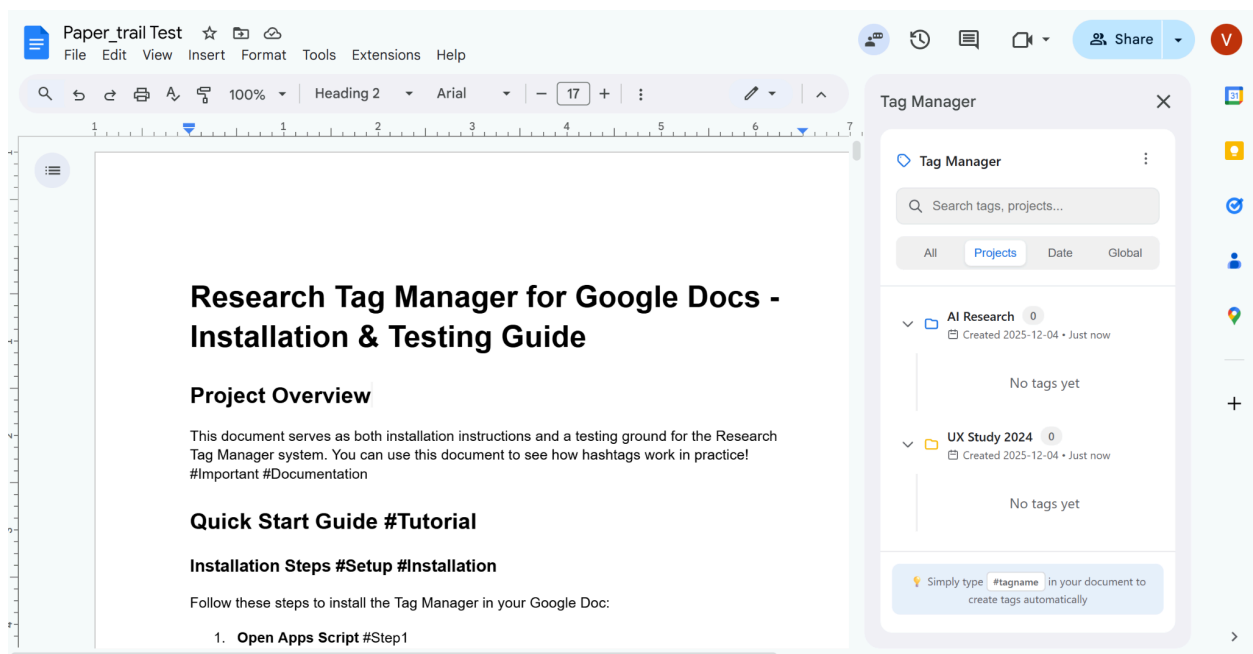


Figure 1. The system employs a three-panel layout with a fixed-height header (48px), flexible main content area, collapsible right sidebar (360px width), and optional footer (32px). The sidebar can be toggled to maximize document viewing space

2.1.1 Screen Areas Description

Top Navigation Bar:

- A thin bar at the top of the webpage that will stay at a fixed height on the screen, as seen in Figure 1. This nav bar will contain system branding (left), view mode switchers (center-left), and a keyboard shortcut button (right).

Smart Sidebar:

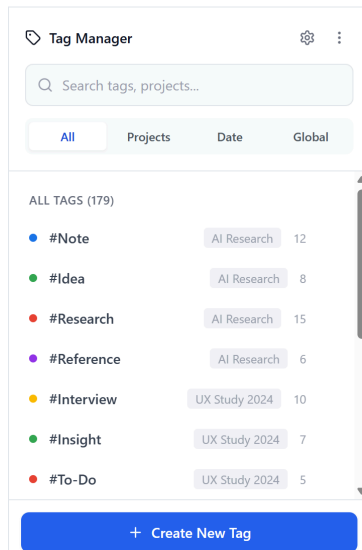


Figure 2. A sidebar that will appear to the right of a document being edited in real time. Allowing the user to organize tags found on the document, and others that have the relating tags as well

- This sidebar will appear on the right side of the screen when prompted open, moving the document over to the left slightly to make room for it. This bar also has the capability of being hidden (collapsible) to allow for the document content to take the focus. The sidebar will contain its UI elements in a stacked format from top to bottom. First, a title, a search bar, filter notches, which are listed from left to right where the user can filter displayed tags based on type, and, finally, a list of tags that fall under the filter or lack thereof, with scrollable capabilities if the list exceeds sidebar height. See Figure 3 for visual.

2.2 Common Components

2.2.1 Buttons

The system uses four button types as shown in Figure 4.

This design system defines four button styles:

Primary Buttons use a solid background with white text. They have no border, a 40px height, and an 8px corner radius. These are intended for high-priority actions such as creating tags or adding projects.

Secondary Buttons use a background with text. They include a 1px border in, a 36px height, and an 8px radius. These support lower-priority actions like canceling or navigating back.

Icon Buttons have a transparent background with icon color. They have no border, a 32px height, and a 4px radius. They are reserved for tool-related controls such as edit, delete, or settings.

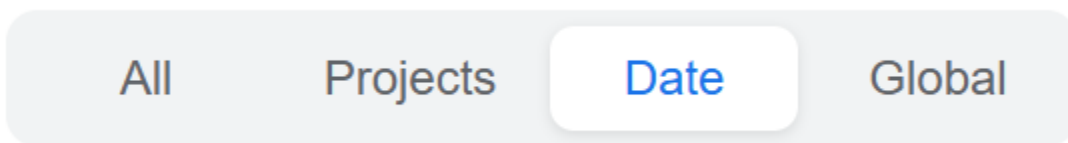


Figure 4. Button specifications define four distinct button types used throughout the system. All buttons shall have hover states with 10% darker background colors and focus states with 2px blue outline .

Button States:

- **Default:** As specified in table above
- **Hover:** Background darkens by 10%, cursor changes to pointer
- **Active/Pressed:** Background darkens by 20%, scale transforms to 98%
- **Disabled:** Opacity set to 40%, cursor changes to not-allowed
- **Focus:** 2px outline in, 2px offset

2.2.2 Input Fields

Element	Specification	Value
Height	Fixed	40px
Border	Default	1px solid
Border	Focus	2px solid
Border Radius	All corners	8px
Background	Default	
Background	Focus	
Padding	Horizontal	12px
Font Size	Body text	14px
Placeholder Color	Gray text	

Figure 5. Input field specifications ensure consistent form element styling. All input fields shall display icons 12px from the left edge when applicable (search icon, calendar icon).

Input Field States:

- **Empty:** Placeholder text visible in #9AA0A6
- **Active/Typing:** Border changes to #1A73E8, background to #FFFFFF
- **Filled:** Text in #202124, background remains #FFFFFF
- **Error:** Border changes to #EA4335, helper text appears below in red
- **Disabled:** Opacity 40%, background #F1F3F4, cursor not-allowed

2.2.3 Icons

The system uses Lucide React icon library with the following standards:

Context	Size	Color	Stroke Width
Navigation Icons	20px	#5F6368	2px
Tag Indicators	16px	Tag color	2px
Metadata Icons	12px	#5F6368	2px
Action Buttons	16px	Inherit	2px
Status Indicators	12px	Status color	2px

Figure 6. Icon sizing and coloring standards maintain visual hierarchy and ensure accessibility. Icons shall always be accompanied by tooltips on hover after 500ms delay.

2.3 Color System

Color Name	Hex Value	Usage
Primary Blue	#1A73E8	Primary actions, active states, links
Success Green	#34A853	Success messages, positive indicators
Warning Yellow	#FDBC04	Warning messages, caution states
Error Red	#EA4335	Error messages, delete actions
Accent Purple	#9334E9	Accent elements, special tags
Info Cyan	#0891B2	Informational messages, highlights
Gray 900	#202124	Primary text, headings
Gray 600	#5F6368	Secondary text, icons
Gray 300	#DADCE0	Borders, dividers
Gray 100	#F1F3F4	Subtle backgrounds
Gray 50	#F8F9FA	Section backgrounds
White	#FFFFFF	Main background, cards

Figure 7. The color palette follows Google's Material Design principles to ensure accessibility and visual consistency. All color combinations shall meet WCAG 2.1 AA contrast requirements (minimum 4.5:1 for normal text).

2.4 Typography

Style Name	Font Size	Font Weight	Line Height	Letter Spacing	Usage
Heading 1	32px	700 (Bold)	40px (1.25)	0	Page titles
Heading 2	24px	600 (Semibold)	32px (1.33)	0	Section headers
Heading 3	18px	600 (Semibold)	24px (1.33)	0	Subsection headers
Body Large	16px	500 (Medium)	24px (1.5)	0	Emphasized body text
Body	14px	400 (Regular)	20px (1.43)	0	Standard body text
Body Small	13px	400 (Regular)	18px (1.38)	0	Secondary information
Caption	12px	600 (Semibold)	16px (1.33)	0.5px	Labels, metadata
Overline	11px	600 (Semibold)	16px (1.45)	1px	Category labels (uppercase)

Figure 8. Typography scale ensures consistent text hierarchy throughout the system. The system shall use system fonts: -apple-system, BlinkMacSystemFont, 'Segoe UI', 'Google Sans', 'Roboto', sans-serif.

2.5 Spacing System

The system uses an 8px base unit for all spacing calculations.

Token Name	Value	Common Usage
space-4	4px	Tight gaps between related icons
space-8	8px	Gaps between form elements
space-12	12px	Standard element padding
space-16	16px	Section padding, card spacing
space-24	24px	Large section gaps
space-32	32px	Major section divisions
space-48	48px	Page padding
space-64	64px	Extra large spacing

Figure 9. All spacing in the system shall use multiples of the 8px base unit to maintain consistent rhythm and alignment.

2.6 Navigation Patterns

2.6.1 Primary Navigation

The system uses the following navigation patterns:

Sidebar Navigation:

- User navigates through tag views using four filter tabs: All, Projects, Date, Global
- Clicking a tab changes the content area below while maintaining scroll position
- Active tab displays with white background and blue text (#1A73E8)
- Tab transitions use 200ms ease-in-out animation

Hierarchical Navigation:

- Projects expand/collapse using chevron icons (ChevronRight/ChevronDown)
- Tags within projects expand to show metadata and tagged items
- Maximum nesting depth: 3 levels (Project > Tag > Items)
- Expansion state persists during session

Keyboard Navigation:

- Tab key moves focus through interactive elements in reading order
- Arrow keys navigate within lists
- Enter/Space activates focused buttons or toggles expansion
- Escape closes modals or command palette
- ⌘+K (Ctrl+K on Windows) opens command palette from any screen

2.6.2 Search and Filter

Search Behavior:

- Search field appears at top of sidebar
- Search activates on keystroke (no submit button required)
- Results filter in real-time with 150ms debounce
- Search queries match against tag names, project names, and item titles
- Clear button (X icon) appears when search has content
- Empty search shows all results

Filter Behavior:

- Filters are mutually exclusive (only one view mode active)
- Clicking a filter tab immediately updates content area
- Filter state persists across page refreshes using localStorage
- Active filter indicated by white background and blue text

2.7 Error Handling

2.7.1 Inline Validation

State	Border Color	Helper Text Color	Icon	Behavior
Valid	#34A853	#34A853	Checkmark	Shows after successful input
Invalid	#EA4335	#EA4335	Alert Circle	Shows on blur if invalid
Warning	#FBBC04	#FBBC04	Alert Triangle	Shows for non-critical issues

Figure 10. Validation states provide immediate feedback to users. Error messages shall appear below input fields with 4px top margin and shall describe the specific problem and required correction.

Validation Timing:

- Validation occurs on blur (when user leaves input field)
- For passwords and critical fields, validation also occurs on submit
- Real-time validation activates after first blur event
- Success indicators appear 300ms after valid input

2.7.2 Error Messages

Error messages shall follow this structure:

Format: [Problem] + [Required Action]

Examples:

- "Tag name is required. Please enter a tag name."
- "Tag name already exists. Please choose a different name."
- "Project name must be 3-50 characters. Please adjust the length."

Error Message Display:

- Position: Directly below the related input field
- Font: 12px regular, color #EA4335
- Icon: Alert circle icon 12px, aligned left
- Animation: Fade in over 150ms
- Duration: Remains visible until error corrected

2.7.3 System-Level Errors**Toast Notifications:**

- Appear in top-right corner, 16px from edges
- Width: 360px fixed
- Auto-dismiss after 5000ms (errors), 3000ms (success)
- User can manually dismiss using X button
- Maximum 3 toasts stacked vertically

Error Types:

- **Connection Error:** "Unable to connect to server. Please check your internet connection."
- **Save Error:** "Failed to save changes. Please try again."
- **Delete Error:** "Cannot delete tag. Tag is currently in use."
- **Permission Error:** "You do not have permission to perform this action."

2.8 Responsive Behavior

Breakpoint	Width	Sidebar Behavior	Layout Changes
Desktop Large	$\geq 1440\text{px}$	Fixed visible 360px	Full three-panel layout
Desktop	1024-1439px	Fixed visible 320px	Sidebar width reduces
Tablet	768-1023px	Overlay drawer	Sidebar overlays on toggle
Mobile	$< 768\text{px}$	Full-screen modal	Sidebar takes full viewport

Figure 11. The system adapts to different viewport sizes while maintaining core functionality. Sidebar behavior changes at defined breakpoints to optimize screen real estate.

Responsive Rules:

- Minimum supported width: 360px
- At tablet size and below, sidebar toggles via hamburger menu
- Font sizes shall not scale below defined minimums
- Touch targets shall be minimum 44x44px on mobile devices
- Hover states replaced with active states on touch devices

2.9 Loading States

Component Loading:

- Skeleton screens display during initial data load
- Skeleton matches layout of actual content
- Loading duration $> 500\text{ms}$ triggers skeleton display
- Spinner icon (20px) used for button loading states

Page Loading:

- Full-page loader for navigation between major sections
- Progress bar at top of viewport (3px height, #1A73E8)
- Indeterminate animation for unknown duration
- Minimum display time: 500ms to prevent flashing

2.10 Accessibility Standards

The system shall comply with WCAG 2.1 Level AA standards:

Keyboard Accessibility:

- All interactive elements accessible via keyboard
- Focus indicators visible with 2px outline
- Logical tab order following reading flow
- Skip links provided for main content areas

Screen Reader Support:

- All images include alt text
- Form inputs associated with labels
- ARIA labels on icon-only buttons
- Status messages announced with aria-live regions

Color Contrast:

- Text contrast minimum 4.5:1 for normal text
- Large text (18px+) minimum 3:1 contrast
- Interactive elements minimum 3:1 contrast
- Error states distinguishable without color alone

3. User Interface Walkthrough

This section provides a comprehensive walkthrough of the Research Tag Manager interface, including navigation flows and detailed screen descriptions. The system centers around the Smart Sidebar interface with multiple view modes for tag organization.

3.1 Navigation Diagram

Figure 12 illustrates the primary navigation paths through the Research Tag Manager system.

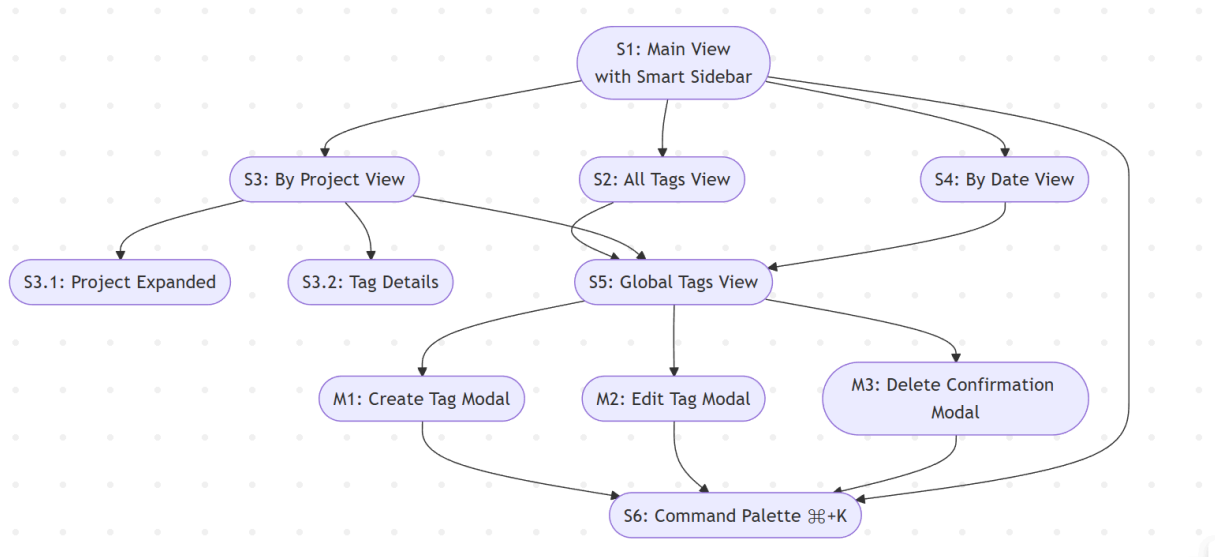


Figure 12. The navigation diagram shows the primary user flows through the Research Tag Manager. Users start at the Main View (S1) and can navigate to four different view modes via filter tabs. Each view provides access to modals for tag management actions (M1-M3). The Command Palette (S6) is accessible from any screen using the keyboard shortcut $\text{⌘}+\text{K}$ (Ctrl+K on Windows).

Navigation Flow Description:

The system employs a single-page application architecture with state-based view switching. Users begin at the Main View (S1), which displays the document area and Smart Sidebar. From S1, users navigate to different organizational views using the filter tab bar:

- **All Tags View (S2):** Displays a flat list of all tags across projects
- **By Project View (S3):** Shows hierarchical project-based organization
- **By Date View (S4):** Organizes tags chronologically
- **Global Tags View (S5):** Shows system-wide priority and status tags

Within the By Project View (S3), users can expand projects (S3.1) to reveal project-specific tags, which can be further expanded to show detailed metadata (S3.2). All views provide access to modal dialogs for creating (M1), editing (M2), and deleting (M3) tags. The Command Palette (S6) can be invoked from any screen using the keyboard shortcut.

3.2 Screen Descriptions

3.2.1 [S1] Main View with Smart Sidebar

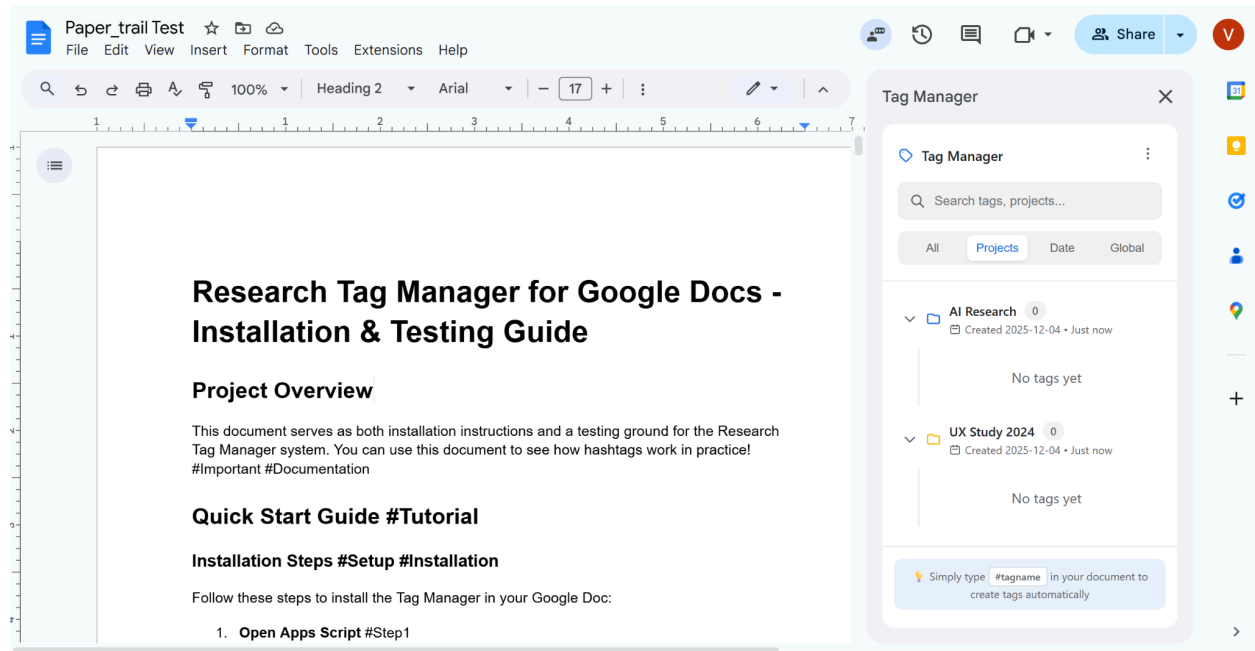


Figure 13. The Main View presents the primary working interface with the document editing area occupying the left portion and the Smart Sidebar fixed to the right edge. The sidebar provides tag management functionality without interrupting document workflow.

Screen Elements:

A. Top Navigation Bar:

- **Location:** Fixed at top of viewport, full width
- **Height:** 48px
- **Elements:**
 - System branding (left): Tag icon + "Research Tag Manager" text
 - "Enhanced" badge in blue (#1A73E8)
 - Keyboard shortcut button (right): Shows "⌘ K" with gray background

B. Document Area:

- **Location:** Left side, flexible width
- **Contents:**
 - Project indicator: Folder icon + "AI Research" in blue
 - Document title: "Neural Network Architecture Study" (32px, bold)
 - Metadata row: Last edited date, author name, inline tag chips
 - Document content: Rich text with inline tag references
 - Tagged sections: Highlighted blocks with left border accent

C. Smart Sidebar:

- **Location:** Right edge, 360px width
- **Sections (top to bottom):**
 - Header: "Tag Manager" title with Settings and More icons
 - Search bar: Search icon + "Search tags, projects..." placeholder
 - Filter tabs: Four tabs (All, Projects, Date, Global)
 - Content area: Displays current view based on active filter
 - Footer: Two action buttons (Create New Tag, Add Project) + keyboard hint

Navigation Actions:

- Click filter tabs to switch between view modes (navigates to S2-S5)
- Click "⌘ K" button or press keyboard shortcut to open Command Palette (S6)
- Click Settings icon to access system preferences
- Click tag items in sidebar to filter document content

User Interaction Flow:

1. User views document content in main area
2. User searches for tags using search bar
3. User switches between view modes using filter tabs
4. User clicks specific tags to highlight related content in document
5. User creates new tags using "Create New Tag" button (opens M1)

3.2.2 [S2] All Tags View

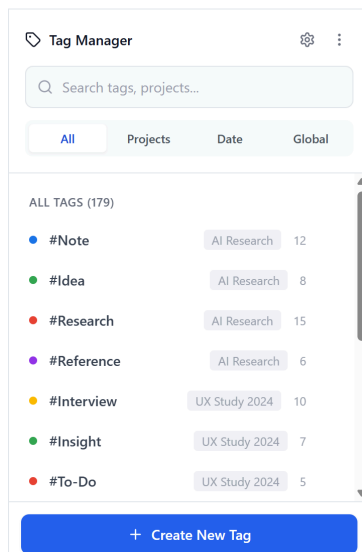


Figure 14. The All Tags View presents a comprehensive flat list of every tag in the system, regardless of project or date. Each tag displays its associated project as a badge and usage count.

Screen Elements:

Sidebar Content Area:

- **Header:** "All Tags (65)" - shows total tag count
- **List Structure:** Vertical scrollable list
- **Each Tag Item Contains:**
 - Color indicator dot (8px, left-aligned)
 - Tag name with # prefix (14px, medium weight)
 - Project badge (gray background, small text)
 - Usage count (gray text, right-aligned)
 - Chevron right icon (appears on hover)

Example Tags Displayed:

- #Note | AI Research | 12
- #Idea | AI Research | 8
- #Interview | UX Study 2024 | 10
- #Feature | Product Development | 9
- [Divider line]
- #Important | (global) | 25
- #Urgent | (global) | 12

Interaction Behavior:

- Hover on tag item: Background changes to #F8F9FA, chevron appears
- Click tag item: Filters document to show only items with that tag
- Search field filters list in real-time as user types
- Scroll: Smooth scrolling with visible scrollbar on hover

Navigation:

- Click "Projects" tab to navigate to S3
- Click "Date" tab to navigate to S4
- Click "Global" tab to navigate to S5
- Click any tag to filter document content

3.2.3 [S3] By Project View

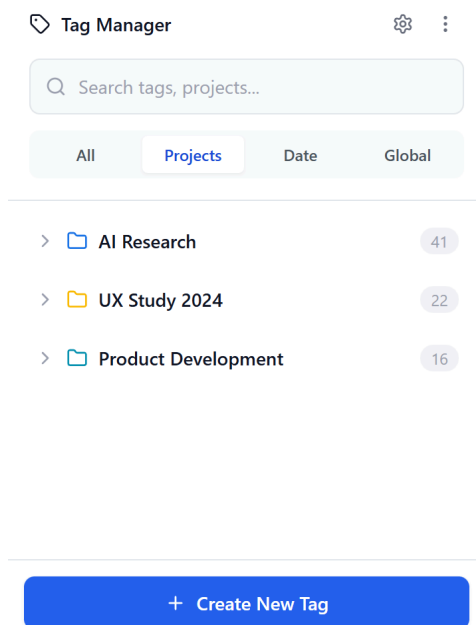


Figure 15. The By Project View organizes tags hierarchically under their parent projects. Users can expand and collapse projects to focus on specific areas of work.

Sidebar Content Area:

- **List Structure:** Hierarchical tree with collapsible nodes

Project Items (Collapsed State):

- Chevron right icon (4px, indicates expandable)
- Folder icon (16px, colored by project)
- Project name (14px, semibold)
- Tag count badge (gray background, shows total tags in project)
- Metadata row (12px, gray text):
 - Calendar icon + "Created [date]"
 - Bullet separator
 - Last activity timestamp

Example Projects:

1. **AI Research** (Blue folder)
 - Created 2024-01-15 • 2 hours ago
 - Badge: 41 tags
2. **UX Study 2024** (Yellow folder)
 - Created 2024-02-01 • 30 mins ago
 - Badge: 22 tags
3. **Product Development** (Cyan folder)
 - Created 2024-01-25 • 4 hours ago
 - Badge: 16 tags

Interaction Behavior:

- Click chevron or project name: Expands project to show tags (navigates to S3.1)
- Hover project: Background changes to #F8F9FA
- Projects remember expansion state during session

Navigation:

- Click collapsed project to expand (navigates to S3.1)
- Click "All" tab to return to S2
- Click "Date" tab to navigate to S4
- Click "Global" tab to navigate to S5

3.2.4 [S3.1] Project Expanded View

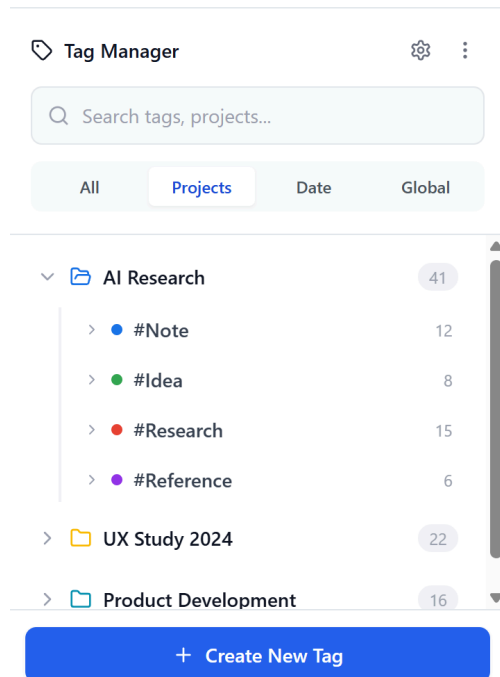


Figure 16. When a project is expanded, the view reveals all tags belonging to that project with a vertical line indicating hierarchy. Each tag can be further expanded to show detailed metadata and tagged items.

Project Header (Same as S3):

- Now shows chevron down icon (indicating expanded state)
- Folder open icon (instead of closed folder)
- Project name, count badge, and metadata row unchanged

Tag List (Indented with left border):

- **Visual Hierarchy:**
 - 24px left margin from project name
 - 2px gray border (#E8EAED) on left side
 - 8px left padding from border

Individual Tag Item:

- Chevron right icon (expandable indicator)
- Color dot (8px, matches tag color)
- Tag name with # prefix (14px, medium)
- Usage count (12px, gray, right-aligned)
- Edit and Delete icons (appear on hover)

Interaction Behavior:

- Click project header: Collapses project back to S3 state
- Click tag chevron: Expands tag to show metadata (navigates to S3.2)
- Hover tag item: Edit and Delete icons fade in
- Click Edit icon: Opens Edit Tag Modal (M2)
- Click Delete icon: Opens Delete Confirmation (M3)

Navigation:

- Click tag to expand metadata (navigates to S3.2)
- Click project header to collapse
- Click different project to expand it and collapse current

3.2.5 [S3.2] Tag Details View

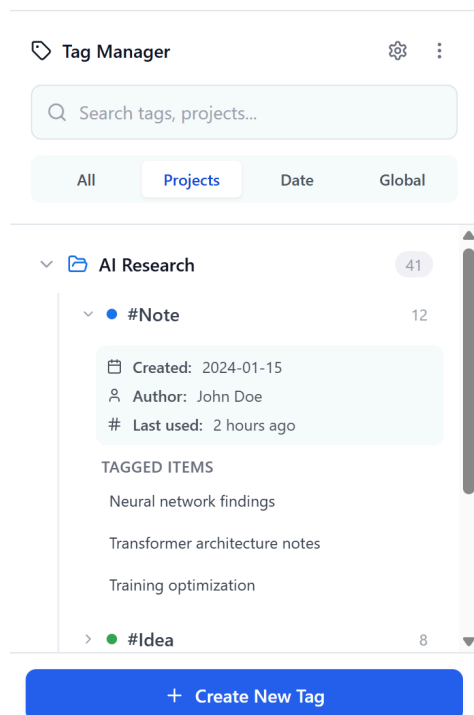


Figure 17. The Tag Details View displays comprehensive metadata about the selected tag including creation information, author, and a list of all items currently tagged with it. This view provides full context for tag usage.

Tag Header:

- Chevron down icon (indicates expanded)
- Color dot indicator
- Tag name with # prefix
- Usage count
- Edit and Delete action icons (visible on hover)

Metadata Panel:

- **Background:** Light gray (#F8F9FA)
- **Border radius:** 8px
- **Padding:** 8px
- **Contents (three rows):**

Row 1 - Created Date:

- Calendar icon (12px, gray)
- "Created:" label (bold, 12px)
- Date value (12px, gray)
- "at" connector
- Time value (12px, blue #1A73E8, bold)
- **Row 2 - Author:**
 - User icon (12px, gray)
 - "Author:" label (bold, 12px)
 - Author name (12px, gray)
- **Row 3 - Last Used:**
 - Hash icon (12px, gray)
 - "Last used:" label (bold, 12px)
 - Relative time (12px, gray)

Tagged Items Section:

- **Header:** "TAGGED ITEMS" (11px, semibold, uppercase, gray)
- **List:** Vertical list of document titles
- **Each Item:**
 - 2px left border (transparent, becomes blue on hover)
 - Title text (12px, gray)
 - Hover state: Background #E8F0FE, border #1A73E8

Interaction Behavior:

- Click tag header: Collapses tag back to S3.1 view
- Click Edit icon: Opens Edit Tag Modal (M2)
- Click Delete icon: Opens Delete Confirmation (M3)
- Click tagged item: Navigates to that item in document
- Hover tagged item: Background highlights, left border appears

Navigation:

- Click tag header to collapse
- Click tagged item to jump to document location
- Click different tag to expand it and collapse current

3.2.6 [S4] By Date View

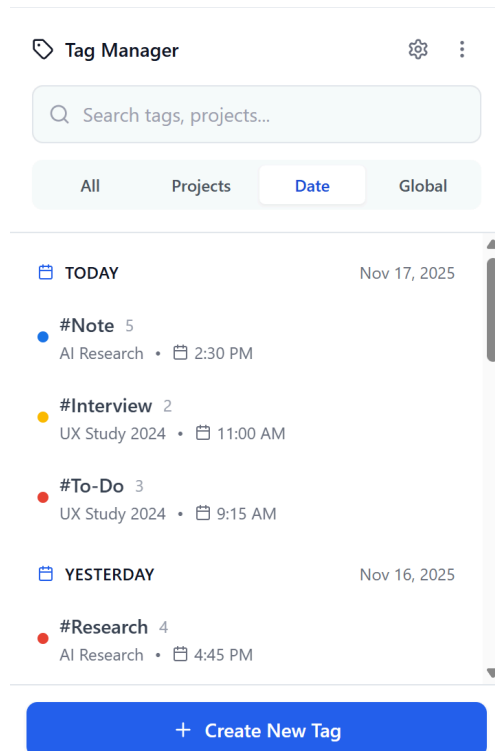


Figure 18. The By Date View organizes tags chronologically into intuitive time periods, allowing users to track recent activity and find tags based on when they were created or last used.

Each section consists of:

- **Sticky Header:**
 - Calendar icon (12px, blue)
 - Period label (11px, semibold, uppercase)
 - Date range (11px, gray, right-aligned)
 - Background: White (stays visible during scroll)
 - Z-index: 10

Time Period Organization:

1. **TODAY** - Nov 17, 2025
2. **YESTERDAY** - Nov 16, 2025
3. **THIS WEEK** - Nov 11-16, 2025
4. **LAST WEEK** - Nov 4-10, 2025
5. **EARLIER THIS MONTH** - Nov 1-3, 2025
6. **OLDER** - Before Nov 2025

Tag Items within Each Period:

- Color dot indicator (8px)
- Tag name and count (first row)
- Project name + timestamp (second row, 11px, gray)
 - Format: "[Project] •  [Time]"
- Chevron right icon (hover only)

Interaction Behavior:

- Scroll: Section headers remain sticky at top
- Hover tag item: Background changes to #F8F9FA, chevron appears
- Click tag: Filters document to show only items with that tag
- Click chevron: Shows quick preview of tag details

Navigation:

- Click "All" tab to navigate to S2
- Click "Projects" tab to navigate to S3
- Click "Global" tab to navigate to S5
- Scroll to view different time periods

3.2.7 [S5] Global Tags View

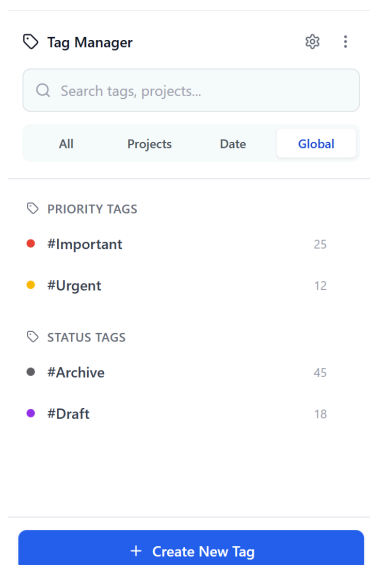


Figure 19. The Global Tags View displays system-wide tags that apply across all projects. Tags are categorized into Priority tags (Important, Urgent) and Status tags (Archive, Draft).

Priority Tags Section:

- **Header:**
 - Tag icon (12px)
 - "PRIORITY TAGS" label (11px, semibold, uppercase)
 - Top margin: 8px

Tag List:

- Each tag displays:
 - Color dot (8px)
 - Tag name with # prefix (14px, medium)
 - Usage count (12px, gray)
 - Chevron right icon (hover only)

Status Tags Section:

- **Header:** Same format as Priority Tags
- **Label:** "STATUS TAGS"
- **Top margin:** 24px from previous section

Tag Details:

- **Important:** Red dot ([#EA4335](#)), for high-priority items
- **Urgent:** Yellow dot ([#FBB03B](#)), for time-sensitive items
- **Archive:** Gray dot ([#5F6368](#)), for completed/archived items
- **Draft:** Purple dot ([#9334E9](#)), for work-in-progress items

Interaction Behavior:

- Hover tag: Background changes to [#F8F9FA](#)
- Click tag: Filters document to show items with that global tag
- Tags can be combined with project-specific tags
- Click chevron: Shows tag usage across all projects

Navigation:

- Click "All" tab to navigate to S2
- Click "Projects" tab to navigate to S3
- Click "Date" tab to navigate to S4
- Click any tag to filter document content

3.2.8 [M1] Edit Tag Modal

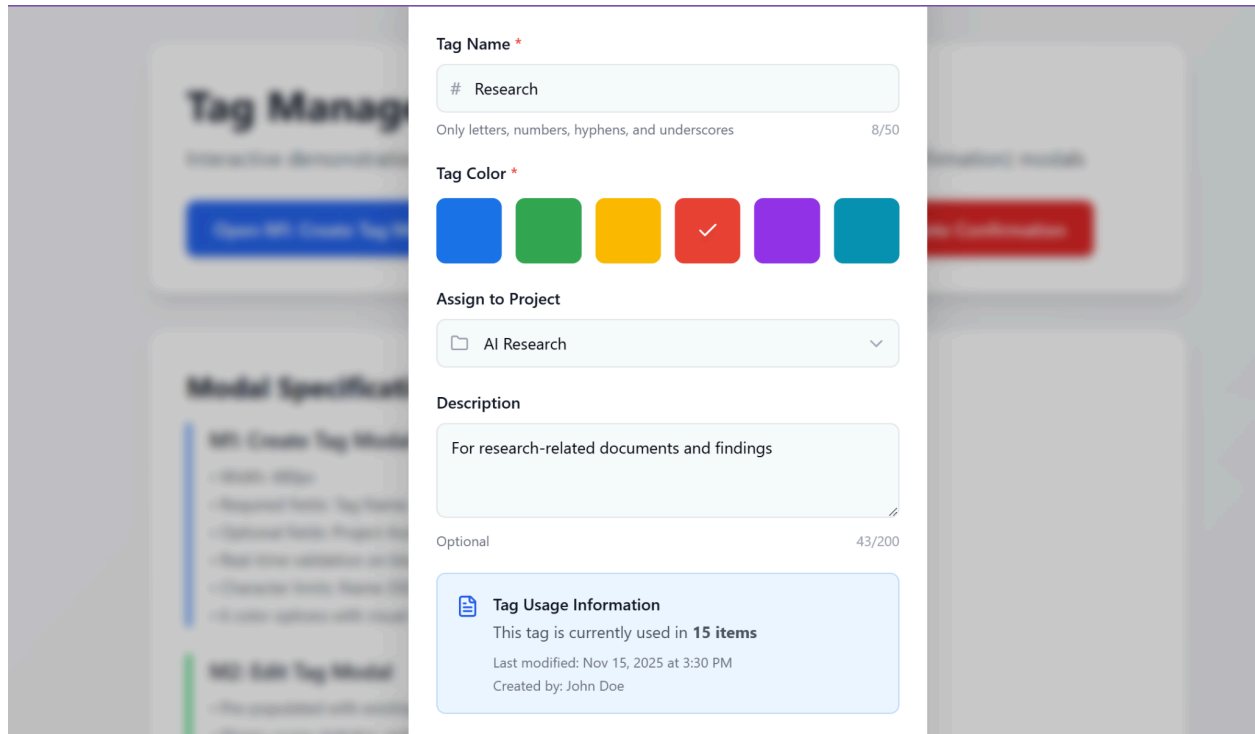


Figure 20. The Edit Tag Modal allows users to modify existing tag properties including name, color, project assignment, and description. All fields are pre-populated with current tag values.

Modal Structure:

- **Overlay:** Full viewport, rgba(0,0,0,0.2) background with 8px blur
- **Modal Container:**
 - Width: 480px
 - Background: White
 - Border radius: 16px
 - Box shadow: 0 20px 60px rgba(0,0,0,0.3)
 - Position: Centered in viewport

Modal Header:

- **Title:** "Edit Tag: #[TagName]" (24px, semibold)
- Example: "Edit Tag: #Research"
- **Close Button:** X icon (top right)

Modal Body:

Field 1 - Tag Name (Required):

- Label: "Tag Name" with red asterisk
- Input field specifications:
 - Placeholder: "Enter tag name (e.g., Research, Note)"
 - Height: 40px
 - Max length indicator: "0/50 characters"
- Prefix indicator: # symbol in gray (automatically prepended)

Field 2 - Color Selection (Required):

- Label: "Tag Color" with red asterisk
- Color palette grid: 6 colors in 2 rows
- Each color swatch:
 - Size: 48px × 48px
 - Border radius: 8px
 - Selected state: 3px border in #1A73E8
 - Hover state: Scale to 105%
- Colors available:
 - Blue (#1A73E8), Green (#34A853), Yellow (#FBBC04)
 - Red (#EA4335), Purple (#9334E9), Cyan (#0891B2)

Field 3 - Project Assignment (Optional):

- Label: "Assign to Project"
- Dropdown select:
 - Placeholder: "Select a project (optional)"
 - Options: List of existing projects + "None (Global Tag)"
 - Height: 40px
 - Chevron down icon on right

Field 4 - Description (Optional):

- Label: "Description"
- Textarea:
 - Placeholder: "Add a description for this tag (optional)"
 - Height: 80px
 - Max length: 200 characters
 - Character counter: "0/200"
 - Resize: Vertical only

Additional Information Panel:

- **Background:** Light blue (#E8F0FE)
- **Padding:** 12px
- **Border radius:** 8px
- **Content:**
 - "This tag is currently used in 15 items"
 - "Last modified: Nov 15, 2025 at 3:30 PM"
 - "Created by: John Doe"

Modal Footer:

- **Buttons (right-aligned):**
 - Cancel button: "Cancel" text
 - Delete button: Secondary style with red text, "Delete Tag"
 - Save button: Primary style, "Save Changes" text

Interaction Behavior:

- Click overlay: Closes modal (with unsaved changes warning if applicable)
- Click X button: Closes modal
- Press Escape key: Closes modal
- Tab through fields: Focus moves in logical order
- Required fields: Show error state if submitted empty
- Create button: Disabled (40% opacity) until Tag Name and Color selected
- Character counters: Update in real-time as user types
- Success: Modal closes, new tag appears in sidebar, success toast appears
- Delete button: Opens Delete Confirmation modal (M2)
- Save button: Validates changes before saving
- If tag name changed: Checks for duplicate names
- Success: Modal closes, tag updates in sidebar, success toast appears
- Cancel: Closes without saving, changes discarded

Validation Rules:

- Tag name cannot be empty
- Tag name must be 1-50 characters
- Tag name cannot contain special characters except hyphen and underscore
- Tag name must be unique within selected project (or globally if no project)
- Color selection is mandatory
- Description limited to 200 characters
- Additional check: If tag name changed, must not conflict with existing tags
- Cannot delete required fields that were previously filled

Navigation:

- Click Save: Validates and updates tag, closes modal, returns to previous view
- Click Delete: Opens Delete Confirmation modal (M2)
- Click Cancel: Closes modal, discards changes, returns to previous view

3.2.9 [M2] Delete Confirmation Modal

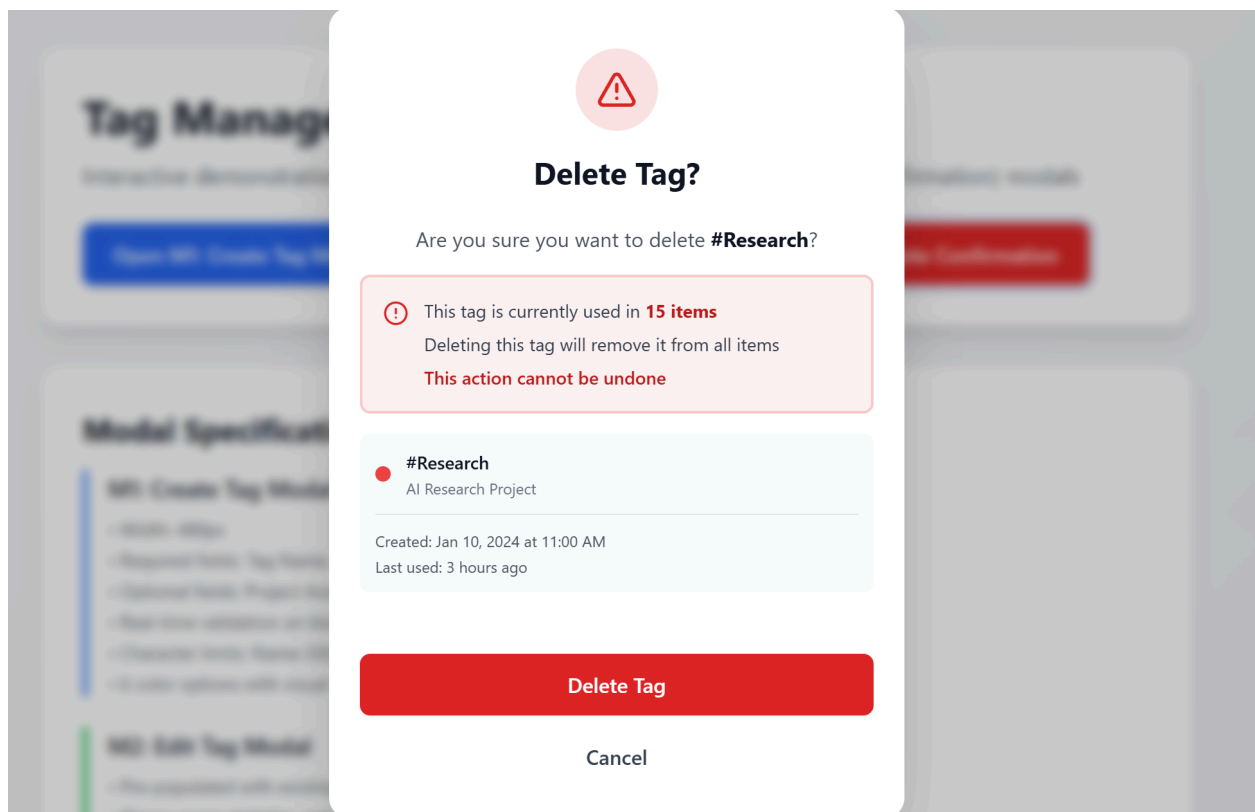


Figure 21. The Delete Confirmation Modal presents a warning dialog before permanently deleting a tag. It shows the impact of deletion and requires explicit confirmation.

Modal Structure:

- **Overlay:** Same as M1 and M2
- **Modal Container:**
 - Width: 400px (narrower than other modals)
 - Background: White
 - Border radius: 16px
 - Centered in viewport

Modal Header:

- **Icon:** Alert triangle in red circle (48px)
- **Title:** "Delete Tag?" (24px, semibold)
- **Padding:** 24px 24px 0 24px
- **Alignment:** Centered

Modal Body (Padding: 24px):

Warning Message:

- Text: "Are you sure you want to delete #[TagName]?"
- Font: 16px, regular, gray (#5F6368)
- Alignment: Centered

Impact Information Panel:

- **Background:** Light red (#FDECEA)
- **Border:** 1px solid #EA4335
- **Border radius:** 8px
- **Padding:** 16px
- **Icon:** Alert circle icon (red)
- **Content:**
 - "This tag is currently used in [X] items"
 - "Deleting this tag will remove it from all items"
 - "This action cannot be undone"
- **Font:** 14px, regular
- **Line spacing:** 1.5

Tag Details:

- **Background:** #F8F9FA
- **Padding:** 12px
- **Border radius:** 8px
- **Content:**
 - Tag name with color dot
 - Project name (if applicable)
 - Creation date
 - Last used date

Modal Footer:

- **Padding:** 0 24px 24px 24px
- **Buttons (full width, stacked):**
 - Delete button:
 - Background: #EA4335 (red)
 - Text: "Delete Tag"
 - Width: 100%
 - Height: 44px
 - Font: 16px, semibold
 - Cancel button:
 - Background: Transparent
 - Text: "Cancel"
 - Width: 100%
 - Height: 44px
 - Font: 16px, medium
 - Top margin: 8px

Interaction Behavior:

- Click Delete:
 - Shows loading spinner on button
 - Deletes tag from system
 - Removes tag from all tagged items
 - Closes modal
 - Shows success toast: "Tag deleted successfully"
 - Returns to previous view with tag removed
- Click Cancel: Closes modal, returns to Edit Tag modal (M2) if came from there, otherwise returns to sidebar
- Click overlay: Same as Cancel
- Press Escape: Same as Cancel
- Focus: Delete button receives focus on modal open (requires deliberate action)

Validation Rules:

- No validation required (confirmation action)
- Delete operation is permanent
- System updates all references immediately

Navigation:

- Click Delete: Executes deletion, closes all modals, returns to main view (S1)
- Click Cancel: Closes this modal, returns to Edit Tag modal (M2) if applicable

3.2.11 [S6] Command Palette

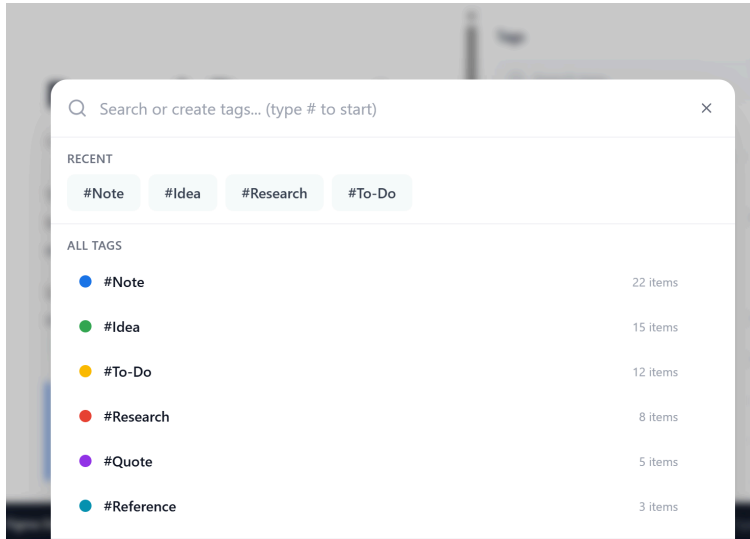


Figure 23. The Command Palette provides keyboard-driven quick access to all tag management functions. It can be invoked from any screen using $\text{⌘}+\text{K}$ (Ctrl+K on Windows) and allows users to search, create, and navigate tags without using the mouse.

Overlay:

- **Background:** `rgba(0,0,0,0.2)` with 8px backdrop blur
- **Coverage:** Full viewport
- **Z-index:** 2000 (above all other elements)

Palette Container:

- **Width:** 640px
- **Position:** Top-center, 128px from top of viewport
- **Background:** White
- **Border radius:** 16px
- **Box shadow:** 0 20px 60px `rgba(0,0,0,0.3)`

Search Header:

- **Padding:** 16px
- **Border bottom:** 1px solid #E8EAED

Search Input:

- **Icon:** Search icon (20px, gray, left-aligned)
- **Placeholder:** "Search or create tags... (type # to start)"
- **Font:** 16px, regular
- **Height:** 48px
- **Border:** None
- **Focus:** Auto-focused on open
- **Close button:** X icon (right-aligned, 16px)

Recent Section:

- **Padding:** 12px 16px
- **Border bottom:** 1px solid #E8EAED
- **Header:** "RECENT" (11px, semibold, uppercase, gray)
- **Tag Pills:** Horizontal scrollable row
 - Each pill:
 - Padding: 8px 16px
 - Background: #F8F9FA
 - Border radius: 8px
 - Hover: Background #E8EAED
 - Font: 14px, medium

Suggestions Section:

- **Padding:** 12px 16px
- **Max height:** 384px
- **Overflow:** Vertical scroll
- **Header:** "ALL TAGS" (11px, semibold, uppercase, gray)

Tag List Items:

- Each item contains:
 - Color dot (12px, left-aligned)
 - Tag name with # prefix (14px, medium)
 - Usage count: "[X] items" (12px, gray)
 - Chevron right icon (16px, right-aligned, visible on hover)
- **Hover state:** Background #F8F9FA
- **Selected state:** Background #E8F0FE
- **Padding:** 12px
- **Border radius:** 8px

Footer:

- **Padding:** 12px 16px
- **Background:** #F8F9FA
- **Border top:** 1px solid #E8EAED

Keyboard Shortcuts Display:

- **Left side:**
 - "↑↓ Navigate" with arrow key icons
 - "↵ Select" with enter key icon
- **Right side:**
 - "Esc Close" with escape key icon
- **Font:** 12px, gray
- **Key indicators:** White background, 1px border, 4px radius

Interaction Behavior:

- **Open:** Press ⌘+K (Ctrl+K) from any screen
- **Type:** Real-time filtering of tag list (150ms debounce)
- **Arrow keys:** Navigate through suggestions (highlights items)
- **Enter key:** Selects highlighted tag or creates new if typed with #
- **Escape key:** Closes palette, returns to previous screen
- **Click outside:** Closes palette
- **Recent tags:** Shows last 4 used tags for quick access
- **Empty search:** Shows all tags
- **Search with #:** Indicates intention to create new tag
 - Footer changes to show "Press Enter to create #[typed_name]"

Search Behavior:

- Matches against tag names, project names
- Case-insensitive search
- Partial matches highlighted in results
- Results sort by relevance, then by usage count
- If search matches no tags: Shows "No tags found. Press Enter to create #[query]"

Navigation:

- Select tag: Closes palette, filters document to selected tag, returns to S1
- Create new tag: Closes palette, opens Create Tag modal (M1) with name pre-filled
- Press Escape: Closes palette, returns to previous screen state
- Click outside: Same as Escape

4. Data Validation

This section specifies all data items that users can enter into the Research Tag Manager system. Each data item is defined with its data type, constraints, format requirements, and validation rules.

4.1 Data Validation Overview

All user input in the Research Tag Manager system undergoes validation at two stages:

1. **Client-side validation:** Occurs in real-time or on blur events to provide immediate feedback
2. **Server-side validation:** Occurs before data persistence to ensure data integrity

Validation errors shall be displayed inline below the relevant input field with descriptive error messages that specify the problem and required correction.

4.2 Tag Management Data Items

Table 1: Tag Creation and Editing Data Fields

Field ID	TG-01	TG-02	TG-03	TG-04
Field Label	Tag Name	Tag Color	Project Assignment	Tag Description
Data Type	String	String (Hex)	String (ID) or NULL	String
Required	Yes	Yes	No	No
Min Length	1	7	0	0
Max Length	50	7	36	200

Format	Alphanumeric, hyphen, underscore only	#RRGGBB	UUID v4 format	Free text
Validation Rules	Must be unique within project scope; Cannot contain spaces; Cannot start with numbers; Case-insensitive comparison	Must be one of six predefined colors: #1A73E8, #34A853, #FBBC04, #EA4335, #9334E9, #0891B2	Must reference existing project ID or be NULL for global tags	No special restrictions; Allows all Unicode characters
Error Messages	"Tag name is required." / "Tag name must be 1-50 characters." / "Tag name can only contain letters, numbers, hyphens, and underscores." / "A tag with this name already exists in this project."	"Please select a color for this tag." / "Invalid color selection."	"Selected project does not exist." / "Invalid project ID format."	"Description must be 200 characters or less."

Table 2: Project Management Data Fields

Field ID	PJ-01	PJ-02	PJ-03
Field Label	Project Name	Project Color	Project Description
Data Type	String	String (Hex)	String
Required	Yes	Yes	No
Min Length	3	7	0
Max Length	50	7	500
Format	Alphanumeric, spaces, hyphens allowed	#RRGGBB	Free text
Validation Rules	Must be unique system-wide; Cannot start or end with spaces; Cannot contain special characters except hyphens and spaces	Must be one of six predefined colors	No special restrictions
Error Messages	"Project name is required." / "Project name must be 3-50 characters." / "A project with this name already exists." / "Project name cannot start or end with spaces."	"Please select a color for this project."	"Description must be 500 characters or less."

Table 3: Search and Filter Data Fields

Field ID	SF-01	SF-02
Field Label	Search Query	Filter Selection
Data Type	String	Enum
Required	No	Yes
Min Length	0	N/A
Max Length	100	N/A
Format	Free text	One of: "all", "projects", "date", "global"
Validation Rules	Accepts all characters; Trims leading/trailing whitespace	Must be one of four predefined values
Error Messages	N/A (Search has no validation errors)	N/A (Radio button selection, cannot be invalid)

4.3 Document Tagging Data Items

Table 4: Inline Tag Application Data Fields

Field ID	DT-01	DT-02	DT-03	DT-04
Field Label	Tag Selection	Tagged Content Start	Tagged Content End	Tagged Content End
Data Type	String (ID)	Integer	Integer	String
Required	Yes	Yes	Yes	Yes
Min Length	36	0	0	1
Max Length	36	2147483647	2147483647	10000
Format	UUID v4	Non-negative integer	Non-negative integer	Free text
Validation Rules	Must reference existing tag ID	Must be valid character position in document; Must be less than end position	Must be valid character position in document; Must be greater than start position	Cannot be empty; Maximum 10,000 characters per tagged selection
Error Messages	"The selected tag does not exist." / "Tag has been deleted."	"Invalid content selection." / "Start position must be before end position."	"Invalid content selection." / "The end position must be after the start position."	"Cannot tag empty selection." / "Tagged selection exceeds a maximum length of 10,000 characters."

4.4 Metadata Data Items

Table 5: System-Generated Metadata Fields

Field ID	Field Label	Data Type	Format	Generated By	Validation Rules
MD-01	Created Date	DateTime	ISO 8601 (YYYY-MM-DDT HH:mm)	System on creation	Cannot be modified by user; Must be valid timestamp
MD-02	Created Time	Time	HH:mm AM/PM	System on creation	Cannot be modified by user; Derived from Created Date
MD-03	Last Modified Date	DateTime	ISO 8601 (YYYY-MM-DDT HH:mm)	System on update	Cannot be manually set; Must be equal to or after Created Date
MD-04	Last Used Date	DateTime	ISO 8601 (YYYY-MM-DDT HH:mm)	System on tag application	Cannot be manually set; Updates when tag applied to content
MD-05	Author	String	Full name or email	System from logged-in user	Cannot be manually changed; Set to current user on creation
MD-06	Usage Count	Integer	Non-negative integer	System calculation	Cannot be manually set; Calculated from number of tagged items
MD-07	Last Activity	String	Relative time (e.g., "2 hours ago")	System calculation	Display-only; Calculated from Last Modified Date

4.5 Validation Rules by Input Type

4.5.1 Text Input Fields

All text input fields shall adhere to the following rules:

Character Encoding:

- System shall accept UTF-8 encoded text
- All Unicode characters are permitted unless specified otherwise
- System shall preserve Unicode characters during save and retrieval

Whitespace Handling:

- Leading and trailing whitespace shall be automatically trimmed on save
- Multiple consecutive spaces shall be preserved within text
- Tab characters shall be converted to single spaces
- Line breaks shall be preserved in multi-line fields

Special Characters:

- HTML special characters (<, >, &, ", ') shall be escaped for display
- SQL injection characters shall be sanitized before database operations
- Script tags shall be stripped from all user input

4.5.3 Numeric Fields

Content Position Fields (DT-02, DT-03):

- Shall accept only non-negative integers
- Shall validate that start position < end position
- Shall validate positions exist within document bounds
- Shall update automatically when document content changes

Usage Count (MD-06):

- Read-only field, user cannot edit
- Shall update in real-time when tags applied or removed
- Shall never be negative
- Shall display 0 for newly created tags

4.6 Validation Timing

Real-Time Validation (During Typing):

- Character count updates on every keystroke
- Maximum length enforcement prevents typing beyond limit
- Format indicators (e.g., # prefix) appear automatically

On-Blur Validation (When Leaving Field):

- Required field check occurs when user tabs out
- Format validation occurs after user finishes typing
- Duplicate name check executes on blur
- Error messages appear below field with red text

On-Submit Validation (When Clicking Save/Create):

- All required fields checked for completion
- All format rules re-validated
- Cross-field validation (e.g., unique names) executed
- Form submission prevented if any validation fails
- Focus moves to first field with error

4.7 Error Message Standards

All error messages shall follow this format:

Structure: [Problem Statement]. [Required Action].

Examples:

- "Tag name is required. Please enter a tag name."
- "Tag name must be 1-50 characters. Current length: 52 characters."
- "A tag with this name already exists. Please choose a different name."

Error Message Display:

- **Position:** Directly below the related input field
- **Font:** 12px, Regular weight
- **Color:** #EA4335 (Error Red)
- **Icon:** Alert circle icon (12px) displayed to left of text
- **Animation:** Fade in over 150ms
- **Duration:** Remains visible until error is corrected
- **Dismissal:** Disappears when user corrects the error

Multiple Errors:

- If multiple fields have errors, all errors display simultaneously
- Focus moves to first field with error on submit
- User can navigate between fields to fix all errors
- Save/Create button remains disabled until all errors resolved

4.8 Data Sanitization

Before data is saved to the database, the system shall apply the following sanitization:

String Fields:

- Trim leading and trailing whitespace
- Remove null bytes (\0)
- Escape HTML special characters for display
- Limit to maximum length specified in validation rules
- Convert to UTF-8 encoding if not already

Tag and Project Names:

- Convert to lowercase for uniqueness comparison
- Store original case for display purposes
- Remove multiple consecutive spaces
- Strip special characters except allowed ones

Free Text Fields (Descriptions):

- Remove HTML tags and script elements
- Preserve line breaks and basic formatting
- Escape quotes and apostrophes for SQL safety
- Maintain Unicode character support

4.9 Data Constraints Summary

Table 6: Data Constraints Summary

Constraint Type	Rule	Applies To	Enforcement
Uniqueness	Tag names must be unique within project scope	TG-01	Database unique constraint + client-side check
Uniqueness	Project names must be unique system-wide	PJ-01	Database unique constraint + client-side check
Required Fields	Must have non-empty value	TG-01, TG-02, PJ-01, PJ-02	Form validation prevents submission
Length Limits	Maximum character count	All string fields	Input maxlength attribute + server validation
Format	Specific pattern match	TG-01, PJ-01	Regex validation on client and server
Referential Integrity	Must reference existing record	TG-03, DT-01	Foreign key constraints in database
Range	Numeric value within bounds	DT-02, DT-03	Comparison validation
Data Type	Must match specified type	All fields	TypeScript typing + database schema

Appendix A – Agreement Between Customer and Contractor

This document represents a formal agreement between the PaperTrail Development Team and Dr. Zihan Wu. The agreement confirms that both parties understand and accept the scope, objectives, and deliverables of the *PaperTrail Project*, which aims to create a document tracking and workflow automation system integrated with Google Docs and Google Drive. Both parties agree to collaborate throughout the project lifecycle, with the development team responsible for implementing the system according to the specifications outlined in this User Interface Design Document, and the customer providing timely feedback and testing support.

In the event of future changes to this document, all revisions must be proposed in writing by either party. Any modification will require a review meeting between the customer and the contractor, and mutual written approval before updates are considered valid. Updated versions of the UIDD will be labeled clearly with version numbers and dates to ensure traceability.

Signatures

Name (Typed)	Signature	Date
Dr. Zihan Wu (Client)	<i>Zihan Wu</i>	12/07/25
Vasu Patel	<i>Vasu Patel</i>	12/05/25
Ben Yandell	<i>Benjamin Yandell</i>	12/05/25
Robert Kulow	<i>Robert Kulow</i>	12/05/25
Arius Ahmad	<i>Arius Ahmad</i>	12/05/25
Anthony Veilleux	<i>Anthony Veilleux</i>	12/05/25

Appendix B – Team Review Sign-off

All members of the PaperTrail Development Team have thoroughly reviewed this User Interface Design Document and agree that it accurately represents the project's goals, scope, and requirements. Each team member confirms that the content and format of this document meet the expectations of both the team and the customer. While minor editorial or stylistic differences may exist, there are no major points of disagreement among the team regarding the material presented herein.

Team Member Signatures

Name (Typed)	Signature	Date
Vasu Patel	<i>Vasu Patel</i>	12/05/25
Ben Yandell	<i>Benjamin Yandell</i>	12/05/25
Robert Kulow	<i>Robert Kulow</i>	12/05/25
Arius Ahmad	<i>Arius Ahmad</i>	12/05/25
Anthony Veilleux	<i>Anthony Veilleux</i>	12/05/25

Appendix C – Document Contributions

The following table identifies each team member's contributions to the creation of this User Interface Design Document. Each member has contributed to writing, editing, and reviewing the document. The percentages represent an estimate of each member's contribution to the total effort.

Team Member	Contributions	Percentage of Work
Vasu Patel	Completed section 2, 3 and 4. Worked on the creation of each figure.	25%
Ben Yandell	Set Up Appendix A, B, C. Section 1.1	20%
Robert Kulow	Oversaw document creation, articulated introduction. Applied format changes for consistency to section 2, 3, & 4.	25%
Arius Ahmad	Updated section 2.1 descriptions and changed formatting to align with previous documents' style.	20%
Anthony Veilleux	Checked for grammatical consistency and fixed minor errors	10%